22AIE205 – Introduction to Python

Class Activity 1

Name: AKEPATI SATHISH REDDY

Reg.NO: CH.SC.U4AIE25001

Department: AIE 25-29

Write a Python program to manage the marks of 10 students using a list.

Your program should:

1. Read the marks of 10 students from the user and store them in a list.

2. Display all the marks.

3. Find and display the highest and lowest marks.

4. Calculate and display the average mark.

5. Count and display the number of students who scored:

o Above the average

o Below 40 (Fail)

6. Display the marks in ascending and descending order.

7. Allow the user to search for a mark and display whether it exists in the list.

8. Replace all marks below 40 with 40 (grace marks) and display the updated list.

9. Display the final list along with the total, average, maximum, and minimum marks.

Constraints:

• Use only Python basics (variables, input/output, operators, conditional statements, loops, and functions) and list operations.

• Do not use tuples, sets, or dictionaries.

• Use appropriate built-in list methods and functions wherever applicable

CODE:

program.py > ...

```python
#CH.SC.U4AIE25001
#INCLASS Activity 1
marks = []
i = 0
print("Enter marks of 10 Students: ")
while(i<10):
    e = int(input(f"Enter marks of student {i+1} : "))
    marks.insert(i,e)
    i+=1
#2
print(f"Marks:{marks}")

#3
high = max(marks)
low = min(marks)
sum_list = sum(marks)

print(f"Highest marks: {high}.")
print(f"Lowest marks: {low}.")
#4
avg = sum_list/10
print(f"Average marks: {avg}")
#5
aboveAvg = 0
countFail = 0

for mark in marks:
    if(mark>avg):
        aboveAvg+=1
    if(mark<40):
        countFail+=1
print(f"No.of students scored above average: {aboveAvg}.")
print(f"No.of students failed: {countFail}.")

#6
tempMarks = marks.copy()
tempMarks.sort()
print(f"Ascending order: {tempMarks}")
tempMarks.reverse()
print(f"Descending order: {tempMarks}")
#7
m = int(input("Enter a mark to search: "))
boo = False
for mark in marks:
    if(m==mark):
        boo = True
        print(f"{m} exists in marks")
if(boo == False):
   print(f"{m} does not exists in marks")
#8
k = 0
while(k<10):
    if(marks[k]<40):
        marks[k] = 40
    k+=1
print(f"Updated Marks: {marks}.")

#9
print(f"Final List: {marks}.")
print(f"Total: {sum(marks)} .")
print(f"Average: {sum(marks)/10}.")
print(f"Maximum: {max(marks)} .")
print(f"Minimum: {min(marks)} .")

```

OUTPUT:

```
PS D:\SEMESTERs\SEMESTER3\22AIE205-Python\290626Class> python -u "d:\SEMESTERs\SEMESTER3\22AIE205-Python\290626Class\program.py"
Enter marks of 10 Students:
Enter marks of student 1 : 12
Enter marks of student 2 : 44
Enter marks of student 3 : 34
Enter marks of student 4 : 56
Enter marks of student 5 : 66
Enter marks of student 6 : 78
Enter marks of student 7 : 88
Enter marks of student 8 : 76
Enter marks of student 9 : 24
Enter marks of student 10 : 10
Marks:[12, 44, 34, 56, 66, 78, 88, 76, 24, 10]
Highest marks: 88.
Lowest marks: 10.
Average marks: 48.8
No.of students scored above average: 5.
No.of students failed: 4.
Ascending order: [10, 12, 24, 34, 44, 56, 66, 76, 78, 88]
Descending order: [88, 78, 76, 66, 56, 44, 34, 24, 12, 10]
Enter a mark to search: 45
45 does not exists in marks
Updated Marks: [40, 44, 40, 56, 66, 78, 88, 76, 40, 40].
Final List: [40, 44, 40, 56, 66, 78, 88, 76, 40, 40].
Total: 568 .
Average: 56.8.
Maximum: 88 .
Minimum: 40 .
```